

ROBOTICS & AI - ADAPTIVE QUADRUPEL ROADMAP

ROBOTICS & AI FULL YEAR DEVELOPMENT ROADMAP

Adaptive Quadruped / Robotics Software / AI Robotics

12 months • 20 hours per week • 52 weeks • 1040 hours

CALENDAR BASELINE

- Start: 12.10.2026. Format of the week: 2 hours theory Monday to Friday and 10 hours practice Sunday; Saturday is free.
- The planned 1040 hours are completed on 19.12.2027, taking into account all vacations.
20-24.12.2027 left as a reserve for final fixes and release checklist.
- Holiday 25.12.2027-02.01.2028 begins after the main completion of the plan.

PURCHASE CHECKLIST AND LINKS

Phase / Gate	Buy or use	Purpose / exit condition	Links
Already owned	MacBook Pro M4; Creality CR-10 SE	The main development computer and FDM prototypes.	-
M1-M2 / Gate 1	Multimeter; 150 mm caliper; M2/M3/M4 fasteners; heat-set inserts; PLA/PETG/TPU	Measurement, assembly and iteration of printed brackets, links and joints.	Rozetka / Prom
M3 / Gate 1	ESP32-S3; optional STM32 Nucleo; breadboard/protoboard; jumper and JST wiring	MCU telemetry, watchdog, encoder and sensor experiments.	Espressif / Prom
M3 / Gate 1	soldering station; regulated fused bench PSU; logic analyzer; optional oscilloscope	Safe on, timing tests and current tests on bench.	Rozetka / Saleae
M3 / Gate 1	IMU; geared DC motor + quadrature encoder; motor driver; current sensor; physical E-stop	One actuator bench with a closed loop position/current/IMU telemetry and a proven safe state.	BNO085 / Pololu / Prom
M6-M8 / Gates 2-3	RGB-D/depth camera; optional 2D LiDAR; onboard Ubuntu-capable compute	Calibration, depth/terrain perception and sim-to-real data path.	Rozetka / Prom
M9-M11 / Gates 3-4	Battery, BMS, fuses, DC-DC; bearings, shafts, couplings and feet	Integration of quadruped without cable with protected power and serviced joints.	Prom / Rozetka
Late Phase / Optional	Jetson or other CUDA compute - only if necessary	Buy only if CUDA perception or learned locomotion really rests on the compute; the M4 Mac remains the main host.	NVIDIA Jetson

Staged purchasing rule

- Do not purchase or build all 12 actuators up front. Validate one actuator bench first, then one 3-DOF leg, and only after successful safety, thermal, encoder, current and control tests scale to the 12-actuator quadruped.
- The final project is an Adaptive Quadruped: 12-DOF, IMU-stabilized body, contact-aware locomotion and balance recovery. Scope is controlled slow locomotion, not Boston Dynamics-style dynamic locomotion.

2. CALENDAR PLAN TO THE END OF 2027

Period	Status	How it is handled
12.10.26-23.12.26	Study	M1-M2: weekday theory 10 hours + practice Sunday 10 hours
24.12.26-03.01.27	Vacation	11 calendar days without training load.
04.01.27-30.04.27	Study	M2-M4: 2 hours of Monday-Friday theory and 10 hours of practice on Sunday
01.05.27-10.05.27	Vacation	10 calendar days without training load.
11.05.27-18.06.27	Study	M5-M6: dynamics, balance control, URDF/Xacro and digital twin.
19.06.27-28.06.27	Vacation	10 calendar days without training load.
29.06.27-09.07.27	Study	M7: calibration, RGB-D/depth and terrain perception.
10.07.27-18.07.27	Vacation	9 calendar days without training load.
19.07.27-20.08.27	Study	M8-M9: state estimation, gait, footholds and recovery.
21.08.27-29.08.27	Vacation	9 calendar days without training load.
30.08.27-24.09.27	Study	M10: learning baseline, classical comparison and OOD fallback.
25.09.27-03.10.27	Vacation	9 calendar days with no study load; 30 scheduled hours missed. hours.
04.10.27-19.12.27	Study	M11-M12: physical integration, HIL, hardening and release.
20.12.27-24.12.27	Buffer	Final polishing, documentation and interview evidence.
25.12.27-02.01.28	Vacation	The roadmap is already complete; full vacation with no deadline shift.

Optional Timing Scenario - Recommended Option

- Start: 12.10.2026. Every week: 2 hours theory Monday to Friday, 10 hours practice Sunday; Saturday is free.
- Additional vacation 25.09-03.10.2027 adds 30 missed hours, with 1040 hours ending 19.12.2027. 20-24.12.2027 remain a flexible reserve; if the start is postponed, reduce the reserve, not vacation periods.

3. CAPSTONE NORTH STAR AND STAGED BUILD

North-Star capstone - Adaptive Quadruped

- A realistic 12-DOF quadruped with active body roll/pitch stabilization from IMU feedback, contact-aware locomotion and balance recovery.
- Target outcomes: stable stand, body attitude compensation, standing on an incline, recovery from repeatable small disturbances, controlled slow gait on flat floor and optional small uneven terrain.
- Explicit non-goal: Boston Dynamics-style dynamic locomotion or a production product claim.

Stage	Definition of done
M2-M3: prove one joint	Printed link/bracket, actuator bench, encoder/current/thermal telemetry, safe limits and physical E-stop.
M4: prove one leg	3-DOF leg with CAD, load cases, FK/IK/Jacobians and repeatable low-speed motion.
M5-M8: prove the model	Rigid-body/contact simulation, IMU and joint fusion, gait state machine and digital twin.
M9-M11: prove the robot	Four legs / 12 actuators, HIL and sim-to-real calibration, classical controller baseline and safety fallback.

4. DEPENDENCY PATH AND STOP CONDITIONS

#	Dependency path	Exit evidence
1	Systems foundation	C++/Python/Rust telemetry, tests, replay and diagnostics.
2	CAD + mechanics + actuation	Printed link/bracket, load cases, FEA note, actuator bench and safety.
3	Leg kinematics + dynamics + control	3-DOF leg, FK/IK/Jacobians, COM/attitude, PID/LQR/MPC and disturbance tests.
4	ROS 2 + digital twin	Quadruped URDF/Xacro, ros2_control, Gazebo contacts/friction/IMU/joints.
5	Perception + state estimation	Calibration, depth/point clouds, IMU+joint+contact fusion and slip awareness.
6	Gait + recovery	Stance/swing state machine or BT, foothold planning, balance recovery and contact-aware safety.
7	Learning + sim-to-real + release	Classical baseline first; RL/IL benchmark, OOD fallback, HIL and release evidence.

Stop conditions

• No scaling past one actuator without current/thermal/E-stop evidence. No four-leg build without one 3-DOF leg passing kinematic and safety checks. No learned policy without a classical controller, fixed metrics and safety/OOD fallback. Nav2, MoveIt and Open-RMF remain breadth labs, not blockers for the quadruped capstone.

5. 12-MONTH MAP

M	Focus	Verified artifact
M1	Systems foundation: C++ + Python + Rust parity	Telemetry core, tests, sanitizer/replay report.
M2	Leg/link/joint/bracket CAD, mechanics, load cases, FEA	Printed single-leg prototype with CAD/BOM, FEA note and fit/deflection test.
M3	Actuator/encoder/current/IMU bench, 3-DOF leg, MCU and safety	One-joint bench, one leg, MCU loop, current/thermal limits and E-stop.
M4	Quadruped body/hip/foot frames, FK/IK/Jacobians	Parametric quadruped CAD and tested leg kinematics library.
M5	Leg/body dynamics, COM/attitude, PID/LQR/MPC, contact control	Disturbance/contact benchmark with classical balance controller.
M6	Quadruped URDF/Xacro, ros2_control, Gazebo digital twin	Contacts, friction, IMU, joints and controller interfaces in simulation.
M7	Calibration, CV, depth/point clouds and terrain perception	Measured RGB-D pipeline; optional 2D LiDAR breadth lab.
M8	IMU+joint+contact fusion, attitude estimation, slip awareness	State-estimation report with dropout and slip metrics.
M9	Gait generation, stance/swing FSM or BT, footholds, recovery	Flat-floor gait and repeatable disturbance/recovery suite; Nav2/MoveIt breadth only.
M10	RL/IL locomotion, domain randomization and push recovery	Policy compared with classical baseline, OOD and safety fallback.
M11	Physical quadruped integration, sim-to-real, HIL and fault injection	12-actuator release candidate with calibration, telemetry and fault matrix.
M12	Hardening, release and interview evidence	Reproducible v1.0, demo, metrics, limitations and setup guide.

Budget invariant

- No extra month or week: the plan remains 48 study weeks x 20 h plus 4 gate weeks x 20 h = 1040 h. M2-M5 redistribute topics inside the existing blocks.

6B. INDUSTRIAL / RT / MULTI-ROBOT / SAFETY LABS

Topic	Block	Mandatory artifact
Custom PCB / KiCad	M3	2-layer sensor/drive interface board: schematic, ERC/DRC, Gerber, BOM.
PREEMPT_RT / cyclicttest	M3, M6	Kernel/options lab, priorities, latency histogram and bounded-allocation note.
EtherCAT / SOEM / IgH	M3, M6	State/PDO loop, timing, induced failure and safe output behavior.
MPC / constrained control	M5	Linear MPC in CasADi/OSQP with limits, timing and infeasibility case.
Open-RMF / multi-robot	M11	Two-robot fleet adapter/schedule scenario with conflict and recovery.
Safety standards	M11	Hazard -> requirement -> test -> evidence traceability using current ISO/IEC references.

Breadth rule

- These labs provide concrete engineering literacy, not professional certification.
- A topic becomes a second-year specialization only after the annual capstone has passed all four gates.

M1. SYSTEMS FOUNDATION: C++ + PYTHON + RUST PARITY

MONTH GOAL
Build a reproducible systems workspace, study lifetime/RAII and diagnostics, and implement the telemetry core and compare C++/Rust/Python boundaries.

VERIFIED ARTIFACT
telemetry-core v1.0 + parser parity report

BUDGET
80 hours

80-HOUR ALLOCATION

Stream	Hours	Share
C++ language and lifetime	26	32%
Build, tests and diagnostics	18	23%
Telemetry + Python/Rust boundaries	28	35%
Evidence package	8	10%

PRACTICAL OUTCOMES

- C++ sensor-frame parser with corrupted-frame tests.
- Bounded ring buffer + deterministic recorder/replayer.
- Rust parity implementation of the parser and queue.
- ASan/UBSan/TSan report; p50/p95/p99 benchmark.
- Python visualization plus pybind11/PyO3/process-boundary comparison.
- Green CI on GCC, Clang and Rust stable.

SOFTWARE AND ENVIRONMENTS

- Ubuntu 24.04
 - GCC/Clang
 - CMake + Ninja
 - GoogleTest/Benchmark
- Rustup/Cargo
 - uv/pytest/NumPy
 - pybind11/PyO3/maturin

M1. RESOURCES AND CRITERIA

MONTH MAP

No	Focus	Hours
1	Baseline, workstation and build model	20 h
2	Object lifetime, ownership and STL	20 h
3	CMake, testing, Linux and diagnostics	20 h
4	Telemetry core + Python/Rust boundary	20 h

REFERENCES, EXERCISES, AND OFFICIAL DOCUMENTATION

• C++ Core Guidelines • LearnCpp • CMake Tutorial • GoogleTest	• The Rust Programming Language • Rustlings • PyO3 guide • Exercism C++ Track
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FORMAL ASSESSMENT / CERTIFICATION

- No mandatory certification. In M1, portfolio evidence is more important than a language certificate.

EXIT CRITERION

- **Explain lifetime/Rail/ownership, localize UB and race, reproduce input stream, and**
Show the measured boundary benchmark.

M1. WEEKS 1-2

Week 1. Baseline, workstation and build model

20 h

- Target-role specification: Robotics Software / Autonomy Engineer.
- Competency matrix: systems, math, hardware, ROS, autonomy, ML.
- Ubuntu 24.04 workstation; native / VM / remote Linux topology.
- Monorepo, ADR, issue templates, experiment log, portfolio index.
- Compilation pipeline: preprocessing, translation units, object files, linking.
- Symbols, linkage, static/shared libraries, ABI awareness.
- CI skeleton: build → lint → test → sanitizer → benchmark artifact.

Week 2. Object lifetime, ownership and STL

20 h

- Fundamental types, references, const correctness, storage duration.
- lvalue/rvalue, move semantics, copy elision, forwarding awareness.
- RAII, deterministic destruction, Rule of Zero / Five.
- Value semantics vs ownership semantics.
- `unique_ptr`, `shared_ptr`, `weak_ptr`; ownership graphs.
- STL containers, iterators, algorithms, ranges, views.
- Error model: exceptions, `error_code`, `std::expected` awareness.
- Templates, concepts and compile-time interfaces.
- Allocator/cache-locality awareness for hot paths.

M1. WEEKS 3-4

Week 3. CMake, testing, Linux and diagnostics

20 h

- Target-based CMake: PUBLIC / PRIVATE / INTERFACE.
- CMake presets, toolchains, install/export, package config.
- GoogleTest fixtures, parameterized tests, death tests.
- ASan, UBSan, TSan, LSan; interpreting reports.
- clang-tidy, clang-format, warnings-as-errors.
- GDB/LLDB, backtrace, core dump, debug symbols.
- Dependency boundaries: apt/rosdep; vcpkg/Conan awareness.
- Linux processes, signals, file descriptors, udev and systemd basics.
- perf/heaptrack workflow and reproducible benchmark conditions.

Week 4. Telemetry core + Python/Rust boundary

20 h

- Binary protocol: framing, endian, checksums/CRC, version fields.
- Ring buffer and bounded queue contracts.
- Deterministic recorder/replayer.
- Monotonic clocks, timestamps, durations, clock conversion.
- Google Benchmark; p50/p95/p99 latency and throughput.
- Python 3.12 + uv, typing, pytest, NumPy visualization.
- pybind11 vs subprocess/IPC vs ROS interface boundary.
- Rust ownership/borrowing, lifetimes, Result, enums/traits, Cargo workspace.
- clippy/rustfmt, cargo test and criterion benchmark.
- PyO3/maturin parity path; compare API, latency and memory with C++.

M2. CONCURRENCY + CAD/FDM + MECHANICAL DESIGN

MONTH GOAL
Study the memory model and bounded pipelines; design a robust parametric assembly with fits, mechanisms, and FDM constraints.

VERIFIED ARTIFACT
telemetry-core v1.1 + modular-sensor-head v1.0

BUDGET
80 hours

80-HOUR ALLOCATION

Stream	Hours	Share
Concurrency and performance	28	35%
Parametric CAD	24	30%
Assemblies, mechanics and FDM	24	30%
Evidence package	4	5%

PRACTICAL OUTCOMES

- Three producer/consumer implementations and contention benchmark.
- Race, deadlock and false-sharing reproduction labs.
- Three fit coupons: holes/shafts, heat-set inserts and bearing seat.
- 6-10 part parametric sensor head with moving joint and cable path.
- Fusion primary model + FreeCAD parity part + Onshape assembly note.
- Printed assembly, measured fit table, drawing, exploded view and BOM.

SOFTWARE AND ENVIRONMENTS

- Fusion 360
 - FreeCAD
 - Onshape
 - OrcaSlicer/PrusaSlicer
- Caliper
 - CR-10 SE
 - perf/Valgrind/heaptrack

M2. RESOURCES AND CRITERIA

No	Focus	Hours
1	C++ concurrency primitives	20 h
2	Bounded pipelines and performance	20 h
3	Parametric CAD: Fusion, FreeCAD, Onshape	20 h
4	Mechanics for robotics: statics, stress/strain, beams, FOS, basic FEA	20 h

REFERENCES, EXERCISES, AND OFFICIAL DOCUMENTATION

- MIT 2.001/2.003 mechanics notes; statics and free-body diagrams.
- Stress/strain, axial load, shear, bending moment and beam deflection.
- Factor of safety and load-case assumptions; SimScale or FreeCAD FEM basics.
- Keep CAD, load-case spreadsheet, FEA mesh/BC screenshot and measured deflection in repo.

EXIT CRITERIA

- One statics/load-path report, one beam-bending calculation, one basic FEA comparison and one printed/assembled fit test. Report predicted vs measured deflection and explain model limits.

M2. WEEKS 1-2

Week 1. C++ concurrency primitives

20 h

- Process, thread, task; concurrency vs parallelism.
- C++ memory model, data race, happens-before.
- `std::thread` / `std::jthread` / `stop_token`.
- `mutex`, `scoped_lock`, `shared_mutex`, lock ordering.
- `condition_variable`, semaphore, latch, barrier.
- Atomics and memory ordering: relaxed/acquire/release/seq_cst.
- False sharing, cache lines, NUMA awareness.
- Thread-safe ownership and cancellation contracts.

Week 2. Bounded pipelines and performance

20 h

- Producer/consumer, backpressure, bounded channels.
- Lock-free / wait-free terminology and trade-offs.
- Scheduling, priority inversion, real-time-ish constraints.
- Latency tails, jitter, deadline misses.
- `perf`, Valgrind, heaptrack, flamegraph workflow.
- Stress tests, property tests, fuzzing awareness.
- Benchmark isolation, warm-up, CPU affinity awareness.
- Contention, queue-depth and backpressure experiment matrix.

M2. WEEKS 3-4

Week 3. Parametric CAD + design-for-test - 20 h

- Design intent, constraints, assemblies, joints, clearances and interference.
- Select load cases for sensor mount, arm bracket or wheel carrier.
- Create drawing/BOM with datum scheme and tolerance assumptions.
- Artifact: revisioned CAD + drawing + fit/interference test.

Week 4. Robotics mechanics lab - 20 h

- Statics: free-body diagrams, reactions, torque paths and equilibrium checks.
- Stress/strain: axial/shear/bending; beam bending and deflection estimate.
- Factor of safety for one load case; state material, print orientation and uncertainty.
- Basic FEA: mesh/BC sanity, convergence spot-check, compare FEA to hand calculation.
- Artifact: mechanics notebook + FEA screenshots/data + printed deflection test + failure note.

M3. EMBEDDED + ACTUATION + PCB + RT/INDUSTRIAL I/O

MONTH GOAL
Safely control an actuator, read signals, build an MCU/RTOS loop, and complete custom PCB, PREEMPT_RT, EtherCAT, and micro-ROS as practical labs.

VERIFIED ARTIFACT
drive-module v1.1 + PCB/RT/fieldbus evidence

BUDGET
80 hours

80-HOUR ALLOCATION

Stream	Hours	Share
Electronics and PCB	20	25%
MCU/RTOS and buses	20	25%
Motors, encoders and instrumentation	20	25%
RT Linux, EtherCAT and integration	16	20%
Evidence package	4	5%

PRACTICAL OUTCOMES

- 6 Wokwi labs plus one real ESP32 motor/encoder bench.
- Port one firmware slice to STM32 and compare peripherals/build.
- Capture and annotate at least three oscilloscope traces.
- Produce KiCad schematic/PCB, ERC/DRC report, Gerber and BOM.
- Run cyclictst before/after tuning; document latency histogram.
- Run SOEM/IgH state/PDO loop and inject disconnect or missed cycle.
- micro-ROS agent reconnect and safe-stop test.

SOFTWARE AND ENVIRONMENTS

- ESP-IDF
 - STM32CubeIDE/PlatformIO
 - FreeRTOS
 - Wokwi
 - KiCad
- PulseView/oscilloscope
 - rt-tests/cyclictst
 - SOEM or IgH EtherLab
 - micro-ROS

M3. RESOURCES AND CRITERIA

MONTH MAP

Nº	Focus	Hours
1	Electronics foundation + custom PCB	20 h
2	ESP32, STM32, FreeRTOS, buses and micro-ROS	20 h
3	Motors, encoders, oscilloscope and characterization	20 h
4	Host/firmware, RT Linux, EtherCAT, safety	20 h

REFERENCES, EXERCISES, AND OFFICIAL DOCUMENTATION

• ESP-IDF Programming Guide • STM32 documentation • FreeRTOS documentation • Wokwi ESP32 simulator • micro-ROS tutorials	• KiCad documentation • Linux PREEMPT_RT documentation • rt-tests / cyclictst • SOEM EtherCAT master • IgH EtherLab EtherCAT
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FORMAL ASSESSMENT / CERTIFICATION

No certification required. Gate 1 requires bench, PCB evidence, measured RT latency and safe failure behavior.

EXIT CRITERION

Bench survives disconnect/restart/stall; limits and safe state checked physically; PCB passes ERC/DRC
RT and EtherCAT results are reproducible.

M3. WEEKS 1-2

Week 1. Electronics foundation + custom PCB

20 h

- Voltage, current, resistance, power, energy.
- Ohm and Kirchhoff laws; series/parallel circuits.
- Digital logic levels, open-drain, pull-up/pull-down.
- Decoupling, filtering, grounding, noise and shielding.
- Level shifting, reverse-polarity and ESD protection.
- Power/current budget, regulator efficiency, thermal awareness.
- Connectors, wire gauge, fuse placement, emergency stop topology.
- Schematic capture, ERC, component footprints and PCB stack-up.
- 2-layer motor/encoder/sensor interface board; DRC, Gerber, BOM and assembly notes.

Week 2. ESP32, STM32, FreeRTOS, buses and micro-ROS

20 h

- ESP32 architecture, boot, flash, logging, partition awareness.
- GPIO, PWM, ADC; calibration and sampling limits.
- UART, I²C, SPI, CAN: framing, arbitration, failure modes.
- Interrupts, debounce, timers, DMA awareness.
- FreeRTOS tasks, queues, semaphores, event groups.
- Watchdog, brownout, reset reason, safe-state state machine.
- Firmware configuration, versioning and reproducible build.
- Minimal STM32 port of one sensor/encoder task.
- micro-ROS agent, transport, executor and safe disconnect behavior.

M3. WEEKS 3-4

Week 3. Motors, encoders, oscilloscope and characterization

20 h

- DC/BLDC/stepper/servo taxonomy and use cases.
- H-bridge and motor driver interfaces.
- Torque-speed-current curve and stall current.
- Gear ratio, backlash, friction, deadband.
- Quadrature encoder, CPR/PPR, direction, wraparound.
- Sampling rate, debounce/filtering, velocity estimation.
- Step response, rise time, overshoot, steady-state error.
- Oscilloscope trigger/coupling/bandwidth; capture PWM, encoder and current ripple.
- Servo/stepper/BLDC taxonomy with cases and failure modes.

Week 4. Host/firmware, RT Linux, EtherCAT, safety

20 h

- Versioned host/firmware protocol.
- CRC, timeout, retry, idempotency, sequence numbers.
- Calibration data and persistent configuration.
- Current, speed, acceleration and travel limits.
- Watchdog and emergency stop behavior.
- Fault injection: disconnect, stall, packet loss, noisy encoder.
- Recorded telemetry and diagnostic fault codes.
- PREEMPT_RT kernel option, rt-tests/cyclictst, priorities and bounded allocation.
- EtherCAT slave discovery, PDO/SDO, distributed clocks and state machine.
- SOEM or IgH lab: cyclic loop, induced loss, watchdog and safe output state.

GATE 1: SOFTWARE + ONE-ACTUATOR / ONE-LEG FOUNDATION

Gate 1 - staged hardware proof

- MacBook Pro M4 and Creality CR-10 SE are already available. Bench tools and safety hardware are acquired before powered experiments.
- One actuator bench passes encoder, current, thermal, watchdog, limit and physical E-stop tests.
- One printed 3-DOF leg is assembled only after the bench evidence is green.

Scale rule

- Do not scale to 12 actuators until one leg demonstrates repeatable low-speed motion, load-case sanity, current limits, safe stop and recoverable faults.

M4. LINEAR ALGEBRA + QUADRUPEL KINEMATICS

Week	Focus	Artifact
1	Linear algebra, calculus and ODE foundation	ODE notebook with Euler/RK4 convergence and unit-aware tests.
2	SO(3)/SE(3), attitude, body frames and quadruped geometry	Frame/quaternion library and body/leg transform tests.
3	Leg FK/IK and Jacobians	3-DOF leg solver with numerical-vs-analytic Jacobian checks.
4	Quadruped body, hip and foot frames	Parametric frame model, workspace/foot reachability plots and review.

M4 acceptance

- The deliverable is a quadruped kinematic model, not a differential-drive or serial-arm capstone. Verify frame conventions, singularities, foot workspace, body attitude coordinates and FK/IK/Jacobian residuals.

M4. KINEMATICS FOR THE QUADRUPEd

Week	Focus	Artifact
1	Linear algebra, calculus and ODE foundation	ODE notebook with Euler/RK4 convergence and unit-aware tests.
2	SO(3)/SE(3), attitude, body frames and quadruped geometry	Frame/quaternion library and body/leg transform tests.
3	Leg FK/IK and Jacobians	3-DOF leg solver with numerical-vs-analytic Jacobian checks.
4	Quadruped body, hip and foot frames	Parametric frame model, workspace/foot reachability plots and review.

M4 acceptance

- The deliverable is a quadruped kinematic model, not a differential-drive or serial-arm capstone. Verify frame conventions, singularities, foot workspace, body attitude coordinates and FK/IK/Jacobian residuals.

M4. WEEKS 1-2

Week 1. Linear algebra, calculus and ODE foundation - 20 h

- Vectors, matrices, eigen/SVD, conditioning, units and frames.
- Derivatives, gradients, Jacobians; first-order and second-order ODEs.
- IVP/equilibrium, linearization intuition, Euler vs RK4 and step-size error.
- Artifact: notebook for pendulum/motor ODE with analytic-oracle and convergence test.

Week 2. $SO(2)/SE(2)$, frames and differential drive - 20 h

- Coordinate transforms, homogeneous matrices, frame direction and units.
- Differential-drive forward/inverse kinematics and integration.
- Use ODE integration to propagate pose; test drift under step-size changes.
- Artifact: transform/differential-drive library + property-based and numerical tests.

M4. WEEKS 3-4

Week 3. $SO(3)$, $SE(3)$, quaternion, twists and adjoint

20 h

- 3D rotation matrices and $SO(3)$ constraints.
- Axis-angle and exponential/logarithm maps.
- Quaternion representation, normalization, interpolation awareness.
- $SE(3)$ homogeneous transforms.
- Twists, screw axes, exponential coordinates.
- Adjoint transformation.
- Frame expression of velocity, force and covariance.
- Singularity of parameterizations and numerical stability.

Week 4. Serial-arm kinematics, FK/IK and Jacobians

20 h

- Serial chains and product-of-exponentials formulation.
- Denavit–Hartenberg awareness and convention pitfalls.
- Forward kinematics implementation.
- Numerical inverse kinematics.
- Space/body Jacobians.
- Singularities and manipulability intuition.
- Finite-difference derivative verification.
- 2-link and 6-DOF arm models; visualization in Python/RViz-ready frames.

M5. DYNAMICS + CONTROL + STATE ESTIMATION + PLANNING ALGORITHMS

MONTH GOAL
Study dynamics, PID/LQR/MPC, KF/EKF/UKF/particle filters, and classical planning through separate notebooks, constraints and comparative benchmarks.

VERIFIED ARTIFACT
robot-math-lab v2.0

BUDGET
80 hours

80-HOUR ALLOCATION

Stream	Hours	Share
Dynamics	16	20%
Control: PID/LQR/MPC	24	30%
Probability and estimation	20	25%
Planning algorithms	16	20%
Evidence package	4	5%

PRACTICAL OUTCOMES

- 2-link Lagrangian dynamics with energy-conservation tests.
- PID + feedforward comparison on motor/trajectory model.
- LQR controller and closed-loop stability/constraint report.
- Linear MPC in CasADi/OSQP with actuator/state limits.
- KF/EKF/UKF/particle-filter comparison on fixed noisy dataset.
- A*/D* Lite/RRT*/PRM benchmark on shared maps.

SOFTWARE AND ENVIRONMENTS

- Python/NumPy/SciPy/SymPy
 - Modern Robotics library
 - CasADi
- OSQP
 - OMPL
 - Jupyter/Matplotlib

M5. DYNAMICS, BALANCE AND CONTACT CONTROL

Week	Focus	Artifact
1	Rigid-body and leg dynamics; COM, inertia and attitude	Dynamics notebook and parameter sensitivity.
2	PID/LQR/MPC, Lyapunov/nonlinear and attitude tracking	Classical balance controller with roll/pitch RMSE and saturation plots.
3	Impedance/contact control and disturbance recovery	Contact-aware controller with push/recovery metrics.
4	Estimation and planning breadth	IMU/joint/contact estimator; foothold/gait planning benchmark.

M5 acceptance

- Compare classical controllers on a rigid-body/leg model. Include COM motion, body attitude, contact force, current saturation and repeatable small-disturbance recovery. Lyapunov/nonlinear material is evidence-informed engineering depth, not a promise of globally optimal locomotion.

M5. WEEKS 1-2

Week 1. Dynamics and actuation models - 20 h

- Mass, COM, inertia, Newton-Euler/Lagrange and manipulator equation.
- Differential-drive and actuator limits; friction, backlash, deadband and efficiency.
- Artifact: dynamics notebook with numerical integration and parameter sensitivity.

Week 2. Feedback + nonlinear/contact control - 20 h

- Classical PID/feedforward/cascaded loops; state-space, linearization, LQR and MPC.
- Lyapunov: candidate functions, positive definiteness, derivative condition and region-of-attraction intuition.
- Nonlinear control fundamentals: feedback linearization awareness, backstepping concept and gain/saturation limits.
- Force/torque control and compliance; impedance/admittance control for contact task.
- Robust/adaptive awareness: disturbance rejection, uncertainty, parameter adaptation and trade-offs.
- Artifact: 3-controller lab with tracking, disturbance and contact-force plots; stability and failure report.

M5. WEEKS 3-4

Week 3. Probability, Kalman family and particle filters - 20 h

- Gaussian/Bayes/covariance; KF, EKF, UKF and particle filter.
- Tune covariance using measured noise; report NIS/NEES or a justified alternative.
- Artifact: estimator notebook with dropout and outlier failure cases.

Week 4. Planning + integrated control benchmark - 20 h

- Dijkstra/A*/D* Lite, RRT/RRT*, PRM, trajectory optimization and OMPL.
- Benchmark completeness, quality, runtime and control tracking on the same scenario.
- Artifact: planner report plus closed-loop comparison of nominal vs disturbed plant.

M6. ROS 2 PRODUCTION + TF2/URDF + ROS2_CONTROL + DIGITAL TWIN

MONTH GOAL
Build a production-oriented ROS 2 workspace with QoS/lifecycle/tracing, correct TF2/URDF, ros2_control, and multiple simulation backends.

VERIFIED ARTIFACT
resilient-robot-stack v1.0

BUDGET
80 hours

80-HOUR ALLOCATION

Stream	Hours	Share
ROS graph, interfaces and packages	22	28%
Executors, QoS and lifecycle	18	22%
TF2, URDF and ros2_control	20	25%
Simulation, tracing and parity	16	20%
Evidence package	4	5%

PRACTICAL OUTCOMES

- Mixed C++/Python/Rust custom-interface workspace.
- QoS failure matrix and lifecycle restart test.
- URDF/Xacro + valid TF tree + automated timestamp/frame tests.
- ros2_control mock hardware and differential-drive controller.
- Gazebo digital twin with encoder/IMU/LiDAR/camera and noise.
- Webots or PyBullet parity scenario with reproducibility note.
- Tracing report with end-to-end latency and sim/bench interface parity.

SOFTWARE AND ENVIRONMENTS

- ROS 2 Jazzy
 - colcon/rosdep/vcstool
 - RViz2/rosbag2
 - ros2_control
 - Gazebo Harmonic + ros_gz
- Webots or PyBullet
 - rclrs
 - Foxglove/PlotJuggler
 - ros2_tracing

M6. RESOURCES AND CRITERIA

MONTH MAP

Nº	Focus	Hours
1	ROS 2 graph, DDS/RMW and language-neutral contracts	20 h
2	Executors, QoS, lifecycle and fault-tolerant components	20 h
3	TF2, URDF/Xacro, SRDF and ros2_control	20 h
4	Observability, Gazebo and simulator comparison	20 h

REFERENCES, EXERCISES, AND OFFICIAL DOCUMENTATION

• ROS 2 Jazzy documentation • ROS 2 Jazzy tutorials • ROS 2 examples • TF2 concepts and tutorials • URDF tutorials • ros2_control documentation	• ROS 2 tracing tutorial • ros2-rust / rclrs • Gazebo + ROS installation • ros_gz integration • Webots User Guide • PyBullet documentation
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FORMAL ASSESSMENT / CERTIFICATION

- The Construct ROS 2 Basics/Certification can be attempted after the workspace passes QoS, TF and replay tests.

EXIT CRITERION

Frame conventions, QoS matrix and lifecycle failures explained without guessing; identical interfaces run in mock, Gazebo and second simulator.

M6. WEEKS 1-2

Week 1. ROS 2 graph, DDS/RMW and language-neutral contracts

20 h

- Workspace, package, ament_cmake/ament_python, colcon, rosdep.
- Nodes, graph, names, namespaces and remapping.
- Topics, services, actions: semantic selection.
- msg/srv/action and IDL contracts.
- Parameters and launch files.
- rclcpp and rclpy minimal nodes.
- ros2 CLI, rqt_graph, introspection and bag recording.
- DDS/RMW discovery, domains, namespaces and remapping.
- rclcpp/rclpy/rclrs nodes with the same msg/srv/action contracts.
- rosbag2 recording and reproducible launch topology.

Week 2. Executors, QoS, lifecycle and fault-tolerant components

20 h

- DDS/RMW, discovery, domains, serialization.
- QoS: reliability, durability, history, depth, deadlines, liveliness.
- QoS compatibility and failure-model matrix.
- Executors, callback groups and multithreaded execution.
- Composition and intra-process communication.
- Lifecycle state machine, diagnostics and logging.
- rosbag2 QoS overrides, replay regression, launch_testing.
- QoS matrix tests under packet loss, late joiner, burst and restart.
- Composition, intra-process zero-copy awareness and callback scheduling.

M6. WEEKS 3-4

Week 3. TF2, URDF/Xacro, SRDF and ros2_control

20 h

- TF tree invariants: one parent, acyclic, connected.
- Static vs dynamic transforms.
- Timestamps, buffering, extrapolation and clock skew.
- URDF links, joints, visual/collision/inertial blocks.
- Xacro macros, properties and reusable components.
- robot_state_publisher and joint_state_publisher.
- RViz frame debugging and automated TF tests.
- Controller manager and update loop.
- Hardware interface types and URDF ros2_control tags.
- read → update → write timing.
- Hardware lifecycle and error handling.
- joint_state_broadcaster and diff_drive_controller.
- Mock hardware and controller switching.
- Limits, chained controllers and introspection.
- SRDF planning groups and end-effectors for future MoveIt integration.

Week 4. Observability, Gazebo and simulator comparison

20 h

- SDF vs URDF and model composition.
- Worlds, plugins, sensors and update rates.
- Mass/inertia, contacts, friction and backlash approximation.
- Sensor noise, latency and drop models.
- ros_gz bridge, spawning and ROS launch integration.
- Real-time factor, deterministic seeds and scenario tests.
- SIL → processor-in-loop → HIL structure; ros2_tracing latency.
- ros2_tracing, PlotJuggler/Foxglove dashboards and replay-based regression.
- Same robot task in Gazebo and one of Webots/PyBullet; compare RTF, fidelity and setup.
- rclrs telemetry node and mixed-language workspace version pinning.

GATE 2: KINEMATICS, CONTROL + DIGITAL TWIN

Gate 2 - model and digital twin

- Quadruped body/hip/foot frames, FK/IK/Jacobians, dynamics and body attitude control are tested.
- URDF/Xacro, ros2_control and Gazebo represent joints, contacts, friction, IMU and latency assumptions.

Breadth labs

- Differential-drive, Nav2 and MoveIt may remain as compact labs for ROS/autonomy evidence. They do not define the final capstone or consume the quadruped acceptance gate.

Pass condition

- Model-to-bench discrepancies are measured and documented; classical stand/balance controller is reproducible before perception or learning is promoted.

M7. CAMERA CALIBRATION + CLASSICAL CV + POINT CLOUDS + DEEP PERCEPTION

MONTH GOAL
Build measured perception pipeline: camera/stereo/hand-eye calibration, OpenCV, point clouds detection/segmentation/pose and ROS integration.

VERIFIED ARTIFACT
perception-evidence-pack v1.0

BUDGET
80 hours

80-HOUR ALLOCATION

Stream	Hours	Share
Camera and calibration	20	25%
Classical CV and ROS image pipeline	20	25%
Point clouds and 3D perception	18	22%
Deep perception and datasets	18	23%
Evidence package	4	5%

PRACTICAL OUTCOMES

- Camera intrinsic calibration with residual plots and rejected frames.
- Stereo/depth reconstruction and scale-error benchmark.
- Hand-eye calibration with independent validation target.
- AprilTag/PnP pose estimate from recorded bag.
- Open3D/PCL registration and point-cloud obstacle baseline.
- Detection/segmentation model with dataset card and ROS topic output.

SOFTWARE AND ENVIRONMENTS

- OpenCV
 - Open3D
 - PCL
- image_pipeline/camera_calibration
 - PyTorch
 - rosbag2/image_transport

M7. RESOURCES AND CRITERIA

MONTH MAP

Nº	Focus	Hours
1	Camera models, stereo/depth and hand-eye calibration	20 h
2	OpenCV, features, optical flow and ROS image transport	20 h
3	Point clouds, registration and 3D geometry	20 h
4	Detection, segmentation, pose estimation and dataset engineering	20 h

REFERENCES, EXERCISES, AND OFFICIAL DOCUMENTATION

• OpenCV tutorials • Open3D documentation • Point Cloud Library documentation	• Stanford CS231n • NVIDIA DLI training and certificates
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FORMAL ASSESSMENT / CERTIFICATION

- OpenCV/NVIDIA certificates optional; calibrated data, residuals, accuracy and latency report have priority.

EXIT CRITERION

- Calibration, timestamps, latency and accuracy are documented; perception replays deterministically from bags and includes failure examples.

M7. WEEKS 1-2

Week 1. Camera models, stereo/depth and hand-eye calibration

20 h

- Pinhole model and projection equation.
- Intrinsic matrix and distortion coefficients.
- Extrinsic transforms and frame conventions.
- Calibration target design and data capture.
- Reprojection error and residual analysis.
- Stereo/depth geometry.
- Hand-eye calibration awareness.
- Eye-in-hand vs eye-to-hand $AX=XB$ formulations.
- Exposure, rolling shutter, motion blur and timestamp effects.

Week 2. OpenCV, features, optical flow and ROS image transport

20 h

- Image representation, color spaces and normalization.
- Convolution, smoothing and sharpening.
- Thresholding and morphology.
- Edges, contours and connected components.
- Features, descriptors and matching.
- Optical flow and motion cues.
- Homography, PnP, ArUco/AprilTag pose.
- sensor_msgs/Image and CameraInfo.
- cv_bridge and encoding conversions.
- image_transport and compressed transport trade-offs.
- camera_calibration package workflow.
- Exact/approximate synchronization.
- Exposure, frame rate, timestamp source and latency.
- Bag-based replay and deterministic perception tests.
- Latency measurement from sensor timestamp to perception output.

M7. WEEKS 3-4

Week 3. Point clouds, registration and 3D geometry

20 h

- Point-cloud formats, organized/unorganized clouds and frame transforms.
- Voxel downsampling, passthrough, outlier removal and crop boxes.
- Normals, curvature and local neighborhoods.
- Plane segmentation, clustering and model fitting.
- ICP point-to-point / point-to-plane and convergence criteria.
- Global registration and RANSAC alignment awareness.
- LiDAR-camera projection and colorizing point clouds.
- Open3D vs PCL APIs, performance and ROS message conversion.

Week 4. Detection, segmentation, pose estimation and dataset engineering

20 h

- Depth-to-point-cloud geometry.
- Open3D/PCL data structures.
- Voxel downsampling and outlier removal.
- Plane segmentation and Euclidean clustering.
- Object detection/segmentation baseline.
- Dataset splits, augmentation and leakage prevention.
- Precision/recall/mAP awareness and inference latency.
- CNN and vision-encoder fundamentals; feature pyramids and transformers.
- Object detection boxes, masks/keypoints and NMS.
- 2D/6D pose estimation, PnP baseline and keypoint methods.
- PyTorch/OpenCV model integration and ROS node packaging.

M8. SENSOR FUSION + EKF/UKF + VISUAL/LIDAR SLAM + EVALUATION

MONTH GOAL
Build measured localization stack: wheel odometry, IMU/GNSS models, EKF/UKF fusion, SLAM Toolbox/RTAB-Map and ATE/RPE evidence.

VERIFIED ARTIFACT
localization-evidence-pack v1.0

BUDGET
80 hours

80-HOUR ALLOCATION

Stream	Hours	Share
Odometry and IMU/time	18	22%
Fusion and covariance tuning	20	25%
SLAM front-end/back-end	22	28%
Evaluation and ablations	16	20%
Evidence package	4	5%

PRACTICAL OUTCOMES

- Wheel-odometry calibration and error model report.
- EKF and UKF on the same bag; consistency and sensitivity analysis.
- SLAM Toolbox and RTAB-Map comparison on shared data.
- Two environments and one loop-closure stress case.
- ATE/RPE evaluation under one parameter ablation.
- Localization loss and sensor dropout recovery scenario.

SOFTWARE AND ENVIRONMENTS

- robot_localization
 - SLAM Toolbox
 - RTAB-Map
- evo
 - rosbag2
 - TUM/KITTI/EuRoC datasets

M8. RESOURCES AND CRITERIA

MONTH MAP

Nº	Focus	Hours
1	Wheel odometry, IMU and timestamp discipline	20 h
2	robot_localization, EKF/UKF and observability	20 h
3	Visual and LiDAR SLAM	20 h
4	Quantitative SLAM evaluation and failure analysis	20 h

REFERENCES, EXERCISES, AND OFFICIAL DOCUMENTATION

• robot_localization documentation • SLAM Toolbox • RTAB-Map ROS • evo trajectory evaluation • TUM RGB-D dataset	• KITTI Vision Benchmark Suite • EuRoC MAV dataset • Probabilistic Robotics • State Estimation for Robotics
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FORMAL ASSESSMENT / CERTIFICATION

- Proof = replayable bags, ATE/RPE, covariance rationale and failure analysis.

EXIT CRITERION

Fusion and SLAM rated numerically; frame/covariance/observability choices and failure recovery are reproducible.

M8. WEEKS 1-2

Week 1. Wheel odometry, IMU and timestamp discipline

20 h

- Wheel encoder measurement model.
- IMU frames, axis conventions and units.
- Gyro bias, accelerometer noise and drift.
- Magnetometer caveats and magnetic disturbances.
- Timestamp synchronization and clock skew.
- Wheel encoder measurement model, quantization, slip and wheel-radius calibration.
- Differential-drive odometry and systematic calibration.
- IMU accelerometer/gyroscope models, bias, scale and axis misalignment.
- Time synchronization, clock skew, transport delay and interpolation.
- Covariance fields, units and consistency checks.

Week 2. robot_localization, EKF/UKF and observability

20 h

- Random variables, distributions and expectation.
- Multivariate Gaussian and covariance ellipses.
- Correlation vs causation; covariance frame.
- Bayes rule and recursive estimation.
- Motion and measurement models.
- Kalman predict/update and innovation.
- EKF Jacobians; NIS/consistency awareness.
- robot_localization state vector, frames and configuration model.
- 2D vs 3D mode; differential/relative measurements.
- EKF predict/update and nonlinear measurement models.
- UKF configuration and comparison on the same recorded trajectory.
- Observability of position, velocity, yaw and sensor bias.
- Replay-based parameter sweep and selected configuration rationale.

M8. WEEKS 3-4

Week 3. Visual and LiDAR SLAM

20 h

- Localization vs mapping vs SLAM.
- Occupancy grids and inverse sensor model.
- Scan matching and odometry priors.
- Front-end / back-end split.
- Pose graph and loop closure.
- Graph optimization intuition.
- 2D LiDAR vs visual/depth SLAM.
- SLAM Toolbox; RTAB-Map overview.
- Visual odometry and LiDAR odometry front-end concepts.
- Feature matching, scan matching and motion priors.
- Keyframes, local maps and data association.
- SLAM Toolbox synchronous/asynchronous and localization modes.
- RTAB-Map visual/RGB-D/LiDAR workflows.
- Relocalization, lifelong mapping and map serialization.

Week 4. Quantitative SLAM evaluation and failure analysis

20 h

- Recorded datasets and ground truth alignment.
- ATE and RPE.
- Drift per distance/time.
- Loop-closure residuals and map repeatability.
- Parameter sweep and controlled ablation.
- Calibration/fusion error budget.
- Dataset card and failure taxonomy.
- ATE/RPE, drift per distance/time and loop-closure metrics.
- NIS/NEES, innovation plots and change detection.
- Ground-truth alignment and coordinate/time association.
- Failure taxonomy: bias, timestamp, wheel slip, motion blur and loop closure.

M9. GAIT, FOOTHOLDS, BALANCE RECOVERY + SAFETY

Week	Focus	Artifact
1	Gait generation and stance/swing state machine or BT	Deterministic crawl gait with timing and transition tests.
2	Foothold planning and contact-aware safety	Reachability/foothold planner with collision, force and slip checks.
3	Balance recovery and disturbance tests	Repeatable push tests, peak deviation, settling time and fall/recovery rate.
4	Breadth labs: Nav2 and MoveIt	Small navigation/manipulation lab retained for breadth; no capstone dependency.

M9 acceptance

- The quadruped must stand and execute a controlled slow flat-floor gait in simulation, with body attitude feedback, contact state and a safe stop. No dynamic running or Boston Dynamics-style claim is required.

M9. RESOURCES AND CRITERIA

MONTH MAP

No	Focus	Hours
1	Nav2 architecture, costmaps, planners and controllers	20 h
2	BehaviorTree.CPP, recovery and safety envelope	20 h
3	Movelt 2: model, planning scene, IK and OMPL	20 h
4	Pick-and-place, task planning and trajectory execution	20 h

REFERENCES, EXERCISES, AND OFFICIAL DOCUMENTATION

• Nav2 documentation • Nav2 First-Time Robot Setup Guide • Nav2 tutorials • BehaviorTree.CPP	• Movelt 2 documentation • Movelt 2 tutorials • Movelt Setup Assistant • OMPL documentation
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FORMAL ASSESSMENT / CERTIFICATION

- The Construct ROS Navigation/ROS Certification may be taken only after both Nav2 and Movelt scenario suites are reproducible.

EXIT CRITERION

- Nav2 and Movelt 2 have measured success, recovery/failure tests and documented tuning; safety envelope is never violated.

M9. WEEKS 1-2

Week 1. Nav2 architecture, costmaps, planners and controllers

20 h

- Lifecycle-managed Nav2 servers.
- BT Navigator, map server and AMCL.
- Costmap2D architecture.
- Robot footprint and collision geometry.
- Obstacle, inflation and voxel layers.
- TF, odometry, laser/depth and timestamp prerequisites.
- First-time robot setup checklist.
- A*, Dijkstra, Theta*, hybrid/state-lattice awareness.
- NavFn and Smac planner families.
- DWB, Regulated Pure Pursuit and MPPI controllers.
- Path smoothing and velocity smoothing.
- Controller frequency and transport delay.
- Goal checker and progress checker.
- Parameter tuning methodology and experiment matrix.
- 20-trial protocol, success rate, time, path length and recovery metrics.

Week 2. BehaviorTree.CPP, recovery and safety envelope

20 h

- BehaviorTree.CPP nodes, ports and blackboard.
- Sequence, fallback, retry, timeout and recovery patterns.
- Spin, back-up, wait and clear-costmap behaviors.
- Blocked path, localization loss, stale sensor, planner timeout.
- Collision Monitor and speed zones.
- Watchdog, emergency stop and degradation modes.
- Custom recovery node and fault-injection tests.
- Navigation-to-manipulation handoff and shared safety state.

M9. WEEKS 3-4

Week 3. MoveIt 2: model, planning scene, IK and OMPL

20 h

- MoveIt architecture: move_group, planning components and interfaces.
- URDF/SRDF and MoveIt Setup Assistant workflow.
- Planning groups, end effectors, virtual joints and disabled collisions.
- Planning scene, collision objects, attached objects and ACM.
- Kinematics plugins, IK solvers, constraints and search-state effects.
- OMPL planning pipelines, planner selection and benchmark setup.
- MoveGroupInterface / MoveItCpp C++ usage.
- RViz interactive planning and state-validity debugging.
- ros2_control integration and simulated joint trajectory controller.

Week 4. Pick-and-place, task planning and trajectory execution

20 h

- Grasp frames, pre-grasp/retreat, approach vectors and object frames.
- MoveIt Task Constructor stages and solution search.
- Cartesian paths, jump threshold and collision checking.
- Trajectory time parameterization, velocity/acceleration limits and smoothing.
- Execution monitoring, controller feedback and abort behavior.
- Grasp failures, collision failures, IK failures and recovery actions.
- Pick-and-place scenario with randomized object poses.
- 20-trial report: success rate, planning time, execution error and failure taxonomy.
- Navigation-to-manipulation handoff as a measured integrated scenario.

GATE 3: QUADRUPED SIMULATION + LOCOMOTION EVIDENCE

Gate 3 - simulated locomotion evidence

- URDF/Xacro + Gazebo digital twin, contact/friction/IMU tests, state estimation and deterministic gait.
- Depth/terrain perception is calibrated; foothold and slip signals are logged.

Acceptance evidence

- Stable stand ≥ 60 s; roll/pitch tracking RMSE; incline target approximately ± 10 -15 deg subject to hardware capability; repeatable small push recovery with peak deviation and settling time; controlled slow flat-floor gait.

M10. LEARNED LOCOMOTION WITH CLASSICAL SAFETY BASELINE

Block	Focus	Artifact
1	Imitation learning and classical baseline	Dataset, behavior cloning baseline and evaluation protocol.
2	RL locomotion and domain randomization	Policy training with randomized friction, mass, delay and sensor noise.
3	Push recovery and OOD detection	Compare policy against PID/LQR/MPC; detect OOD and hand off to safe fallback.
4	Evidence package	Ablation, holdout, failure taxonomy and compute budget; no learned policy overrides safety.

M10 acceptance

- Learning is optional evidence layered on the classical balance/gait controller. The robot must remain within current, thermal, attitude and contact safety bounds, and every policy result must include an OOD/safety fallback.

M10. RESOURCES AND CRITERIA

MONTH MAP

No	Focus	Hours
1	Reproducible PyTorch and robot datasets	20 h
2	Behavior cloning, DAgger, ACT and diffusion policy	20 h
3	RL, offline RL, curriculum and domain randomization	20 h
4	VLA, language grounding and safety gating	20 h

REFERENCES, EXERCISES, AND OFFICIAL DOCUMENTATION

• PyTorch tutorials • LeRobot GitHub • LeRobot documentation • Hugging Face Datasets • OpenAI Spinning Up	• Berkeley CS285 Deep Reinforcement Learning • Stable-Baselines3 documentation • Gymnasium documentation • NVIDIA Isaac Sim documentation • Isaac Lab documentation
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FORMAL ASSESSMENT / CERTIFICATION

- NVIDIA DLI is optional only when Isaac evidence exists; otherwise complete the Modern Robotics credential or skip certification.

EXIT CRITERION

- Every learned policy is compared with classical/BC baseline on fixed holdout; OOD, latency, safety gate and fallback are documented.

M10. WEEKS 1-2

Week 1. Reproducible PyTorch and robot datasets

20 h

- Tensors, devices, dtypes and shapes.
- Autograd and computation graph.
- nn.Module, loss, optimizer and scheduler.
- Dataset/DataLoader and batching.
- Train/eval loop and metrics.
- Checkpointing, seeds and reproducibility.
- Profiler, mixed precision and export awareness.
- Transfer learning and fine-tuning.
- Vision encoders and embeddings.
- Augmentation and domain shift.
- Class imbalance and confidence calibration.
- Dataset versioning and immutable splits.
- Leakage checks and dataset cards.
- Experiment tracking with MLflow/W&B.
- Action/state alignment, timestamping and normalization.

Week 2. Behavior cloning, DAgger, ACT and diffusion policy

20 h

- Demonstrations and action/state alignment.
- Behavior cloning baseline.
- Compounding error and DAgger concept.
- Temporal context and action chunking.
- ACT architecture and evaluation.
- Diffusion policy awareness.
- LeRobotDataset, rollout, train, evaluate.
- Covariate shift, compounding error and closed-loop evaluation.
- Offline-policy checkpoints, deterministic evaluation and confidence intervals.

M10. WEEKS 3-4

Week 3. RL, offline RL, curriculum and domain randomization

20 h

- MDP, state, observation, action, transition, reward and discount.
- Value functions, Q function, policy and advantage.
- On-policy vs off-policy; PPO and SAC practical training runs.
- Exploration, entropy, reward shaping and sparse rewards.
- Offline RL dataset constraints, distribution shift and conservative evaluation.
- CQL/IQL concepts and a small fixed-dataset comparison.
- Curriculum learning and task randomization.
- Domain randomization: mass, friction, latency, noise, lighting and textures.
- Isaac Lab task configuration, parallel environments and GPU constraints.
- MuJoCo comparison: throughput, fidelity and reproducibility.
- Sim-to-real assumptions, system identification and deployment gate.

Week 4. VLA, language grounding and safety gating

20 h

- Multimodal transformers, vision-language representations and language grounding.
- Action tokenization, continuous-action heads and action chunking.
- Fine-tuning, adapters/LoRA and frozen encoder trade-offs.
- Pretrained policy/model as an evaluation workflow.
- Embodied tasks: success, latency, generalization and OOD metrics.
- Closed-loop latency, success, generalization and OOD tests.
- Language ambiguity, prompt/task perturbation and failure taxonomy.
- Policy shield, workspace limits, confidence gate and classical fallback.
- Human override, emergency stop and audit log for learned actions.
- Isaac Sim synthetic-data workflow and one domain-randomization ablation.

M11. ADAPTIVE QUADRUPEL INTEGRATION + SIM-TO-REAL + HIL

M11 capstone integration

- Assemble the physical 12-actuator Adaptive Quadruped only after the staged bench and 3-DOF leg gates are green. Integrate actuator calibration, IMU+joint+contact fusion, gait tuning, HIL and sim-to-real parameter alignment.
- Inject encoder dropout, IMU bias, contact loss, overcurrent, thermal limit and communication faults. Verify physical E-stop and independently verified safe state.

Release candidate artifacts

- One-command simulation and hardware bringup; calibration dataset; controller comparison; telemetry schema; fault matrix; 30-50+ trial log or statistically justified sample size.
- Open-RMF is a short breadth lab only. Jetson is optional and late, only if CUDA perception or learned locomotion is demonstrably needed.

M11. WEEKLY PLAN: PHYSICAL QUADRUPEL

M11 capstone integration

- Assemble the physical 12-actuator Adaptive Quadruped only after the staged bench and 3-DOF leg gates are green. Integrate actuator calibration, IMU+joint+contact fusion, gait tuning, HIL and sim-to-real parameter alignment.
- Inject encoder dropout, IMU bias, contact loss, overcurrent, thermal limit and communication faults. Verify physical E-stop and independently verified safe state.

Release candidate artifacts

- One-command simulation and hardware bringup; calibration dataset; controller comparison; telemetry schema; fault matrix; 30-50+ trial log or statistically justified sample size.
- Open-RMF is a short breadth lab only. Jetson is optional and late, only if CUDA perception or learned locomotion is demonstrably needed.

M11. WEEKLY PLAN: HIL AND SIM-TO-REAL

M11 capstone integration

- Assemble the physical 12-actuator Adaptive Quadruped only after the staged bench and 3-DOF leg gates are green. Integrate actuator calibration, IMU+joint+contact fusion, gait tuning, HIL and sim-to-real parameter alignment.
- Inject encoder dropout, IMU bias, contact loss, overcurrent, thermal limit and communication faults. Verify physical E-stop and independently verified safe state.

Release candidate artifacts

- One-command simulation and hardware bringup; calibration dataset; controller comparison; telemetry schema; fault matrix; 30-50+ trial log or statistically justified sample size.
- Open-RMF is a short breadth lab only. Jetson is optional and late, only if CUDA perception or learned locomotion is demonstrably needed.

M11. WEEKLY PLAN: FAULTS, SAFETY AND TRIALS

M11 capstone integration

- Assemble the physical 12-actuator Adaptive Quadruped only after the staged bench and 3-DOF leg gates are green. Integrate actuator calibration, IMU+joint+contact fusion, gait tuning, HIL and sim-to-real parameter alignment.
- Inject encoder dropout, IMU bias, contact loss, overcurrent, thermal limit and communication faults. Verify physical E-stop and independently verified safe state.

Release candidate artifacts

- One-command simulation and hardware bringup; calibration dataset; controller comparison; telemetry schema; fault matrix; 30-50+ trial log or statistically justified sample size.
- Open-RMF is a short breadth lab only. Jetson is optional and late, only if CUDA perception or learned locomotion is demonstrably needed.

M11. RELEASE ARTIFACTS

M11 capstone integration

- Assemble the physical 12-actuator Adaptive Quadruped only after the staged bench and 3-DOF leg gates are green. Integrate actuator calibration, IMU+joint+contact fusion, gait tuning, HIL and sim-to-real parameter alignment.
- Inject encoder dropout, IMU bias, contact loss, overcurrent, thermal limit and communication faults. Verify physical E-stop and independently verified safe state.

Release candidate artifacts

- One-command simulation and hardware bringup; calibration dataset; controller comparison; telemetry schema; fault matrix; 30-50+ trial log or statistically justified sample size.
- Open-RMF is a short breadth lab only. Jetson is optional and late, only if CUDA perception or learned locomotion is demonstrably needed.

M12. RESOURCES AND CRITERIA

MONTH MAP

No	Focus	Hours
1	Test matrix, stress, performance and compatibility	20 h
2	Reproducible release and evidence package	20 h
3	Technical interview loops	20 h
4	Credential, applications and final gate	20 h

REFERENCES, EXERCISES, AND OFFICIAL DOCUMENTATION

• ROS 2 Quality Guide • Robotics interview questions • The Construct ROS Certification Exam • Modern Robotics Specialization - Coursera	• Autodesk Certified Associate in CAD for Mechanical Design • NVIDIA DLI training and certificates • ROS 2 Lyrical documentation
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FORMAL ASSESSMENT / CERTIFICATION

- Choose at most one credential. Portfolio release, tests and technical evidence remain the priority.

EXIT CRITERION

- An unfamiliar engineer reproduces the system; you can defend architecture, hardest failure, safety, metrics, RT/industrial and learning trade-offs.

M12. WEEKS 1-2

Week 1. Test matrix, stress, performance and compatibility

20 h

- Unit, integration, launch, replay and scenario tests.
- Deterministic environment and pinned dependencies.
- Regression thresholds and golden metrics.
- Configuration/hyperparameter freeze.
- Benchmark protocol and measurement conditions.
- Confidence intervals and controlled ablations.
- Known limitations and unresolved failure modes.
- Stress/soak tests, memory growth, handle leaks and watchdog behavior.
- Failure detection/recovery matrix and stale-state verification.
- Latency/jitter, CPU/GPU/memory/bandwidth and queue-depth profiles.
- Power/thermal profile and resource-limit enforcement.
- ROS 2 Lyrical / Gazebo Jetty compatibility sandbox; core remains pinned to Jazzy/Harmonic.

Week 2. Reproducible release and evidence package

20 h

- README: problem, acceptance criteria and architecture.
- Install, bringup, troubleshooting and recovery guide.
- BOM, CAD exports, firmware, configs and calibration files.
- System context and timing/data-flow diagrams.
- Licenses and third-party notices.
- GitHub Release and reproducible artifact bundle.
- 3-minute demo video and 10-minute technical deep dive.
- Firmware binaries/source, flashing guide and calibration/config files.
- Dataset card, model card and evaluation protocol.
- Architecture, timing/data-flow and safety diagrams.
- Troubleshooting, failure-recovery guide and maintenance checklist.

M12. WEEKS 3-4

Week 3. Technical interview loops

20 h

- Coding: arrays, graphs, geometry, concurrency, tests.
- C++ lifetime, RAII, UB, race, performance trade-offs.
- Python profiling and native/process boundary.
- Frames, quaternions, FK/IK, Jacobian, EKF, PID.
- ROS graph, DDS/QoS, executors, TF2, lifecycle and deployment.
- Calibration, SLAM, Nav2, control and failure analysis.
- Robotics system design and project deep dive.
- MoveIt, Open-RMF, RT Linux, fieldbus, safety and VLA trade-offs.
- System-design interview with faults, observability, rollback and safe stop.
- Project deep dive: hardest bug, rejected options, metrics and personal contribution.

Week 4. Credential, applications and final gate

20 h

- CV, GitHub, LinkedIn and project index alignment.
- Six technical STAR stories.
- Five mock interviews.
- Twenty targeted applications/referrals.
- One upstream issue, reproduction or pull request.
- Optional certification exam.
- Six-month post-roadmap gap analysis.
- Choose one: ROS certification, Modern Robotics, Autodesk CAD or NVIDIA DLI.
- Credential only after capstone release candidate is complete.
- Six robotics-focused CV/GitHub/LinkedIn alignments to target role and ownership.
- 20 targeted applications/referrals and role-specific project framing.
- Post-roadmap 90-day specialization plan and open-source contribution.
- Publish v1.0 release and retrospective.

GATE 4: HIRE-READY ADAPTIVE QUADRUPEL RELEASE

Metric	Acceptance target
Stable stand	>=60 s without fall; report body attitude and joint/current telemetry.
Attitude tracking	Roll/pitch target tracking RMSE, peak error and settling time reported.
Incline	Adjustable target approximately +/-10-15 deg, subject to hardware capability and explicit limit.
Disturbance recovery	Repeatable small push; peak deviation, settling time, fall/recovery rate and failure taxonomy.
Locomotion	Controlled slow gait on flat floor; uneven terrain is optional and clearly labeled.
Contact and slip	Foot-slip distance/rate, contact confidence and false-contact events.
Safety and health	Joint/current/IMU/contact telemetry; current and thermal limits; physical E-stop and independently verified safe state.
Timing and sample size	Control-loop latency/jitter; 30-50+ trials or statistically justified sample size.
Learning	RL/IL compared with classical baseline; domain randomization, OOD detection and safe fallback.

Release rule

- No claim of dynamic locomotion is made unless separately demonstrated. The v1.0 release must include limitations, hardware capability bounds, reproducible setup and raw trial summaries.

A. PORTFOLIO PROJECTS: EVIDENCE STANDARD

#	Portfolio project	Evidence
1	Systems + embedded telemetry	C++/Python/Rust parity, MCU loop, watchdog, replay and fault logs.
2	Printed quadruped mechanics	CAD, load cases, FEA/hand checks, fit and deflection tests.
3	Actuator and 3-DOF leg bench	Encoder/current/IMU data, calibration, safe limits and E-stop.
4	Quadruped kinematics and balance	FK/IK/Jacobians, COM/attitude, PID/LQR/MPC and disturbance plots.
5	Adaptive Quadruped capstone	12-DOF body, gait, contact-aware state estimation, recovery, sim-to-real, trials and safety case.
6	Breadth labs	Differential-drive/Nav2/MoveIt/Open-RMF samples kept compact and explicitly secondary.

Portfolio standard

- Project #5 is the North-Star outcome. Include the measurable acceptance table, raw telemetry, test protocol, failure taxonomy, classical-vs-learned comparison, HIL evidence and 3-minute demo.

B. HARDWARE STACK AND SAFETY MINIMUM

Layer	Minimum stack
Compute	MacBook Pro M4 for development; Ubuntu-capable onboard compute late in the plan; Jetson optional only for a demonstrated CUDA need.
Fabrication	Creality CR-10 SE; PLA/PETG/TPU; heat-set inserts; M2/M3/M4 fasteners; bearings, shafts, couplings and feet.
Bench	Multimeter, caliper, fused bench PSU, soldering station, logic analyzer and optional oscilloscope.
Actuation	Geared DC motor + quadrature encoder, driver, current sensor, IMU; scale from one actuator to 12 only after Gate 1.
Safety	Physical E-stop, fuses, BMS, current/thermal limits, watchdog and independently verified safe state.
Perception	RGB-D/depth camera; optional 2D LiDAR; calibration and synchronization evidence.
Software	ROS 2 Jazzy, Gazebo Harmonic, quadruped URDF/Xacro, ros2_control, rosbag2 and HIL tooling.

Safety minimum

• Power budget, fusing, current and thermal derating, joint travel/speed limits, physical E-stop, software safe state and fault-injection evidence are required before untethered operation.

C. CERTIFICATION MAP			
Priority	Credential	When	Use case
High	The Construct ROS Certification Exam	After M9	Useful for ROS/Nav2 skills; combine with Nav2 + MoveIt evidence.
High	Modern Robotics Specialization - Coursera	M4-M10	Strong math/kinematics/dynamics/planning structure.
Medium	Autodesk Certified Associate in CAD for Mechanical Design	M12	Relevant for CAD-heavy robotics roles.
Conditional	NVIDIA DLI training and certificates	M10-M12	Only when Isaac Sim/Lab appears in the evidence package.
Conditional	ROS-Industrial training and events	After M9	Good with employer sponsorship or industrial-control target roles.

Official links

- [The Construct ROS Certification Exam](#)
- [Modern Robotics Specialization - Coursera](#)
- [Autodesk Certified Associate in CAD for Mechanical Design](#)
- [NVIDIA DLI training and certificates](#)
- [ROS-Industrial training and events](#)

Selection rule

- Choose at most one credential during the 12-month plan.
- A public capstone with metrics is the primary hiring signal.

D1. COMPLETE TOPIC COVERAGE: SOFTWARE, LANGUAGES AND TOOLING

Topic	Scheduled block	Status
C++ types/references/const/lifetime/RAII/move/smart pointers	M1 W2	Mandatory
STL/ranges/templates/concepts/error handling	M1 W2	Mandatory
CMake install/export and colcon integration	M1 W3, M6 W1	Mandatory
GoogleTest/sanitizers/static analysis/fuzzing	M1 W3	Mandatory
Concurrency/atomics/bounded queues/backpressure	M2 W1-W2	Mandatory
Linux processes/udev/systemd/perf	M1 W3	Mandatory
Python typing/packaging/NumPy/asyncio/multiprocessing/profiling	M1 W4	Mandatory
Rust ownership/lifetimes/traits/Result/Send+Sync	M1 W4	Mandatory
pybind11/PyO3/maturin/process boundary	M1 W4	Mandatory
Mixed-language workspace and rclrs	M6 W1, W4	Mandatory

Coverage rule

- Every row has an implementation/configuration/CAD artifact, benchmark or failure test. Reading alone is not accepted.

D2. COMPLETE TOPIC COVERAGE: CAD, MECHANICS AND MANUFACTURING

Topic	Scheduled block	Status
Fully constrained sketches/parameters/feature history	M2 W3	Mandatory
Assemblies/components/joints/motion/interference	M2 W3	Mandatory
Fasteners/bearings/shafts/couplings/gears/belts/rails	M2 W3	Mandatory
Fits/tolerance stack/datums/basic GD&T	M2 W3	Mandatory
Statics/free-body diagrams/load paths/equilibrium	M2 W4	Mandatory
Stress/strain, shear and beam bending/deflection	M2 W4	Mandatory
Factor of safety + load-case/uncertainty note	M2 W4	Mandatory
Basic FEA: mesh/BC/convergence + hand-check comparison	M2 W4	Mandatory
FDM anisotropy/overhangs/supports/inserts + deflection test	M2 W4	Mandatory
Drawings/BOM/exploded view/STEP-STL-3MF/revision	M2 W3-4, M12 W2	Mandatory

Coverage rule

- Every row has an implementation, configuration/CAD artifact, benchmark or failure test. Reading alone is not accepted; FEA is basic engineering evidence, not a substitute for professional certification.

D3. COMPLETE TOPIC COVERAGE: ELECTRONICS, EMBEDDED AND ACTUATION

Topic	Scheduled block	Status
Voltage/current/power, grounding/noise/ESD	M3 W1	Mandatory
Custom PCB schematic/ERC/DRC/Gerber/BOM	M3 W1	Mandatory
ESP32/STM32 interrupts/timers/DMA	M3 W2	Mandatory
FreeRTOS tasks/queues/semaphores/event groups	M3 W2	Mandatory
UART/I2C/SPI/CAN framing and faults	M3 W2	Mandatory
Motors, drivers, torque-speed/current, encoders	M3 W3	Mandatory
Oscilloscope PWM/encoder/current traces	M3 W3	Mandatory
Host protocol/calibration/limits/fault injection	M3 W4	Mandatory
micro-ROS publisher/agent/reconnect	M3 W2, W4	Mandatory
PREEMPT_RT/rt-tests/cyclicttest/priorities	M3 W4, M6 W4	Mandatory
EtherCAT PDO/SDO/distributed clocks/SOEM or IgH	M3 W4, M6 W4	Mandatory

Coverage rule

- Every row has an implementation/configuration/CAD artifact, benchmark or failure test. Reading alone is not accepted.

D4. COMPLETE TOPIC COVERAGE: MATH, DYNAMICS, CONTROL AND PLANNING

Topic	Scheduled block	Status
Linear algebra/eigen/SVD/least squares/conditioning	M4 W1	Mandatory
Calculus/gradients/Jacobians/ODE integration	M4 W1	Mandatory
First/second-order ODEs, IVP, RK4 and convergence tests	M4 W1	Mandatory
SO(2)/SE(2), frames, differential drive	M4 W2	Mandatory
SO(3)/SE(3), quaternion, twists, adjoint	M4 W3	Mandatory
FK/IK/Jacobians/singularities/manipulability	M4 W4	Mandatory
Mass/inertia/Lagrange/Newton-Euler/friction	M5 W1	Mandatory
PID/feedforward/cascades/saturation/anti-windup	M5 W2	Mandatory
Lyapunov stability + nonlinear control fundamentals	M5 W2	Mandatory
Force/torque, compliance, impedance/admittance	M5 W2	Mandatory
Robust/adaptive control awareness + disturbance test	M5 W2	Mandatory
State-space/controllability/observability/LQR/MPC	M5 W2	Mandatory
Gaussian/Bayes/KF/EKF/UKF/particle filter	M5 W3	Mandatory
A*/D*/Theta*/RRT*/PRM/trajectory optimization	M5 W4	Mandatory

Coverage rule

- Every row has a notebook, test, benchmark or failure report.

D5. COMPLETE TOPIC COVERAGE: ROS 2, SIMULATION AND OBSERVABILITY

Topic	Scheduled block	Status
ROS graph/DDS/RMW/discovery/IDL/serialization/domains	M6 W1	Mandatory
Topics/services/actions/parameters/launch/custom interfaces	M6 W1	Mandatory
Executors/callback groups/composition/intra-process	M6 W2	Mandatory
QoS reliability/durability/history/deadline/liveliness	M6 W2	Mandatory
Lifecycle/diagnostics/logging/restart/degradation	M6 W2	Mandatory
TF2 timestamps/buffering/static-dynamic frames	M6 W3	Mandatory
URDF/Xacro/SRDF/robot_state_publisher	M6 W3	Mandatory
ros2_control hardware interfaces/controller manager	M6 W3	Mandatory
rosbag2/replay/launch_testing/tracing/PlotJuggler/Foxglove	M6 W4	Mandatory
Gazebo/SDF/physics/sensors/ros_gz	M6 W4, M11	Mandatory
Webots/PyBullet comparison	M6 W4	Mandatory
SIL/PIL/HIL and sim-real fidelity budget	M6 W4, M11	Mandatory

Coverage rule

- Every row has an implementation/configuration/CAD artifact, benchmark or failure test. Reading alone is not accepted.

D6. COMPLETE TOPIC COVERAGE: PERCEPTION, LOCALIZATION AND SLAM

Topic	Scheduled block	Status
Camera model/distortion/intrinsics/extrinsics	M7 W1	Mandatory
Stereo/depth/hand-eye calibration	M7 W1	Mandatory
OpenCV features/optical flow/RANSAC/PnP	M7 W2	Mandatory
ROS image transport/sync/timestamps/replay	M7 W2	Mandatory
Point clouds/Open3D/PCL/ICP/registration	M7 W3	Mandatory
Detection/segmentation/pose/dataset engineering	M7 W4	Mandatory
Wheel odometry/IMU bias/time synchronization	M8 W1	Mandatory
robot_localization EKF/UKF/observability/covariance	M8 W2	Mandatory
Visual/LiDAR SLAM front/back-end/loop closure	M8 W3	Mandatory
SLAM Toolbox/RTAB-Map	M8 W3	Mandatory
ATE/RPE/NIS/NEES/ablation/error budget	M8 W4	Mandatory

Coverage rule

- Every row has an implementation/configuration/CAD artifact, benchmark or failure test. Reading alone is not accepted.

D5. COMPLETE TOPIC COVERAGE: QUADRUPEd CAPSTONE

Topic	Scheduled block	Status
Quadruped URDF/Xacro, ros2_control and Gazebo contacts	M6	Mandatory
Body attitude, COM, FK/IK/Jacobians and gait state machine	M4-M5/M9	Mandatory
IMU+joint+contact fusion and foot-slip metrics	M8-M9	Mandatory
RGB-D calibration and terrain/depth perception	M7	Mandatory
Nav2 / MoveIt / differential-drive labs	M4/M9	Breadth
Open-RMF / fleet lab	M11	Breadth
RL/IL, domain randomization and OOD fallback	M10	Conditional
Physical E-stop, safe state, current/thermal limits	M3/M11	Mandatory

Coverage rule

- Every mandatory row produces a test, benchmark, CAD/mechanics artifact or failure report. Breadth labs are intentionally smaller and do not displace the quadruped acceptance criteria.

D6. BREADTH TOPICS: NAV2, MOVEIT AND OPEN-RMF

Topic	Scheduled block	Status
Quadruped URDF/Xacro, ros2_control and Gazebo contacts	M6	Mandatory
Body attitude, COM, FK/IK/Jacobians and gait state machine	M4-M5/M9	Mandatory
IMU+joint+contact fusion and foot-slip metrics	M8-M9	Mandatory
RGB-D calibration and terrain/depth perception	M7	Mandatory
Nav2 / Moveit / differential-drive labs	M4/M9	Breadth
Open-RMF / fleet lab	M11	Breadth
RL/IL, domain randomization and OOD fallback	M10	Conditional
Physical E-stop, safe state, current/thermal limits	M3/M11	Mandatory

Coverage rule

- Every mandatory row produces a test, benchmark, CAD/mechanics artifact or failure report. Breadth labs are intentionally smaller and do not displace the quadruped acceptance criteria.

D7. CAPSTONE EVIDENCE AND REVIEW QUESTIONS

Topic	Scheduled block	Status
Quadruped URDF/Xacro, ros2_control and Gazebo contacts	M6	Mandatory
Body attitude, COM, FK/IK/Jacobians and gait state machine	M4-M5/M9	Mandatory
IMU+joint+contact fusion and foot-slip metrics	M8-M9	Mandatory
RGB-D calibration and terrain/depth perception	M7	Mandatory
Nav2 / MoveIt / differential-drive labs	M4/M9	Breadth
Open-RMF / fleet lab	M11	Breadth
RL/IL, domain randomization and OOD fallback	M10	Conditional
Physical E-stop, safe state, current/thermal limits	M3/M11	Mandatory

Coverage rule

- Every mandatory row produces a test, benchmark, CAD/mechanics artifact or failure report. Breadth labs are intentionally smaller and do not displace the quadruped acceptance criteria.

F1. FINAL READINESS CHECKLIST: ADAPTIVE QUADRUPEL CORE

Area	OK	Verification
Staged build	☐	One actuator bench, one 3-DOF leg, then 12 actuators; each gate has measured evidence.
Mechanics	☐	CAD, load cases, FEA/hand checks, fit/deflection and serviceability.
Balance control	☐	Stand, roll/pitch RMSE, incline, disturbance recovery and saturation metrics.
Gait/contact	☐	Stance/swing, footholds, contact confidence, foot slip and safe recovery.
State estimation	☐	IMU+joint+contact fusion, dropout/bias tests and latency/jitter report.
Safety	☐	Physical E-stop, independently verified safe state, current/thermal limits and fault injection.
Trials	☐	30-50+ trials or justified sample size with raw logs and failure taxonomy.
Learning	☐	Classical baseline, RL/IL comparison, domain randomization, OOD and fallback.

Core release gate

- A topic without an artifact remains open. Breadth labs in Nav2, MoveIt and Open-RMF support employability but do not replace quadruped evidence.

F2. FINAL READINESS CHECKLIST: BREADTH, PRODUCTION AND CAREER

Area	OK	Verification
Manipulation	☐	Movelt planning scene, IK, OMPL, MTC pick-and-place and 20+ trials.
Robot learning	☐	Versioned dataset; baseline/ablation/holdout/OOD; safety-gated policy.
Industrial / RT	☐	PCB package, cyclicttest/PREEMPT_RT and EtherCAT PDO/SDO/fault evidence.
Fleet / multi-robot	☐	Open-RMF two-robot task, traffic conflict and recovery report.
Safety engineering	☐	Hazard -> requirement -> test -> evidence matrix using relevant ISO/IEC concepts.
Production	☐	Observability, fault injection, rollback, resource budget and reproducible release.
Career	☐	Capstone release, 3-min demo, 10-min deep dive, 5 mocks and targeted applications.

Capstone release checklist

- Clear task and measurable success criterion.
- Architecture and timing/data-flow diagrams.
- Full BOM, CAD exports, firmware, config and calibration files.
- One-command or well-documented sim and hardware bringup.
- Unit + integration + replay + scenario + HIL/stress tests.
- 20-50+ trials with success rate and confidence interval or justified sample size.
- Latency, jitter, CPU/GPU, memory, bandwidth, power and thermal profile.
- At least eight failure modes with detection/recovery.
- Physical emergency stop / software safe stop.
- 3-minute video and 10-minute technical deep dive.

F3. NORTH-STAR OUTCOME: SAFE, MEASURABLE QUADRUPEd

Outcome area	Evidence
Stable body	>=60 s stand; body roll/pitch tracking RMSE; incline approximately +/-10-15 deg subject to capability.
Locomotion	Controlled slow flat-floor gait; optional uneven terrain; no dynamic-locomotion promise.
Recovery	Repeatable small disturbances with peak deviation, settling time, falls and recovery rate.
Contact awareness	Foot-slip metrics, contact confidence, current/thermal telemetry and fault cases.
Production quality	Latency/jitter, HIL, sim-to-real, calibration, reproducible setup and safety traceability.
Career signal	Public v1.0 release, raw metrics, demo, limitations, design review and interview deep dive.

North-Star statement

- Build and explain a safe, measurable 12-DOF Adaptive Quadruped that can stand, stabilize its body, recover from small repeatable disturbances and walk slowly under contact-aware control. Everything else is supporting breadth or a future extension.

G1. RESOURCE INDEX: SYSTEMS LANGUAGES AND TOOLING	
• C++ Core Guidelines • LearnCpp • CMake Tutorial • GoogleTest • Google Benchmark • Clang documentation • The Rust Programming Language • Rustlings	• PyO3 guide • maturin • uv documentation • NumPy documentation • Exercism C++ Track • Exercism Rust Track • Linux perf
Use rule • Prefer official documentation plus one primary textbook/course per phase. • Do not start a second resource until the first produces a measured artifact.	

G2. RESOURCE INDEX: CAD, EMBEDDED AND ROBOTICS MATH

Foundation / mechanics	Control / ODE / robotics
Autodesk Fusion learning	MIT 18.06 Linear Algebra
Fusion assemblies overview	MIT 2.003/2.004 mechanics and dynamics notes
FreeCAD Manual + FEM workbench	Feedback Systems - Åström & Murray
Onshape Assemblies + tolerance options	Modern Robotics + videos
Prusa: modeling with 3D printing in mind	MIT Underactuated Robotics
SimScale/FEM basics or equivalent	Numerical ODE integration: Euler/RK4 notes
ESP-IDF / STM32 / FreeRTOS	State Estimation for Robotics
KiCad + PREEMPT_RT + micro-ROS	Planning Algorithms - Steven LaValle
SOEM EtherCAT master	CasADi / OSQP / OMPL documentation

Use rule

- Prefer official documentation plus one primary textbook/course per phase. Do not start a second resource until the first produces a measured artifact. For FEA/control, document assumptions and model limits explicitly.

G3. RESOURCE INDEX: ROS 2, SIMULATION AND PERCEPTION

• ROS 2 Jazzy documentation • ROS 2 Jazzy tutorials • ROS 2 Lyrical documentation • ROS 2 examples • TF2 concepts and tutorials • URDF tutorials • ros2_control documentation • ROS 2 tracing tutorial • ros2-rust / rclrs • Gazebo documentation • Gazebo + ROS installation • ros_gz integration • ros_gz project template • Webots User Guide	• PyBullet documentation • MuJoCo documentation • OpenCV tutorials • Open3D documentation • Point Cloud Library documentation • Stanford CS231n • robot_localization documentation • SLAM Toolbox • RTAB-Map ROS • evo trajectory evaluation • TUM RGB-D dataset • KITTI Vision Benchmark Suite • EuRoC MAV dataset
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Use rule

- Prefer official documentation plus one primary textbook/course per phase.
- Do not start a second resource until the first produces a measured artifact.

G4. RESOURCE INDEX: AUTONOMY, ROBOT LEARNING, FLEET, SAFETY AND CAREER

• Nav2 documentation • Nav2 First-Time Robot Setup Guide • Nav2 tutorials • BehaviorTree.CPP • MoveIt 2 documentation • MoveIt 2 tutorials • MoveIt Setup Assistant • TurtleBot3 Manual • PyTorch tutorials • LeRobot GitHub • LeRobot documentation • Hugging Face Datasets • OpenAI Spinning Up • Berkeley CS285 Deep Reinforcement Learning • Stable-Baselines3 documentation • Gymnasium documentation • NVIDIA Isaac Sim documentation	• Isaac Lab documentation • ROS 2 Quality Guide • Open-RMF documentation • rmf_demos repository • ISO 3691-4:2023 • ISO 10218-1:2025 • ISO 10218-2:2025 • ISO/TS 15066:2016 • ISO 13849-1:2023 • IEC functional safety / 61508 • Robotics interview questions • The Construct ROS Certification Exam • Modern Robotics Specialization - Coursera • Autodesk Certified Associate in CAD for Mechanical Design • NVIDIA DLI training and certificates • ROS-Industrial training and events
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Use rule

- Prefer official documentation plus one primary textbook/course per phase.
- Do not start a second resource until the first produces a measured artifact.